Networking Task 02: Network Devices & IP Addressing

Objective

The purpose of this task is to understand common network devices, IP addressing concepts, and how data travels within a network.

Part A: Network Devices Research

1. Router

Purpose

A router connects different networks and allows devices to access the internet.

How it Works

It examines destination IP addresses and forwards data packets to the correct network.

Real-World Usage

Home Wi-Fi routers connect laptops, phones, and other devices to the internet.

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2. Switch

Purpose

A switch connects multiple devices within the same local area network (LAN).

How it Works

It uses MAC addresses to send data directly to the intended device.

Real-World Usage

Used in offices, schools, and organizations to connect computers and printers.

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3. Hub

Purpose

A hub connects multiple devices in a network.

How it Works

It broadcasts data to all connected devices regardless of the destination.

Real-World Usage

Older networks used hubs before switches became common.

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4. Access Point

Purpose

An access point provides wireless network connectivity.

How it Works

It connects wireless devices to a wired network through Wi-Fi.

Real-World Usage

Used in homes, offices, colleges, and public places to provide Wi-Fi access.

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5. Firewall

Purpose

A firewall protects networks from unauthorized access and cyber threats.

How it Works

It monitors and filters incoming and outgoing network traffic according to security rules.

Real-World Usage

Used in computers, routers, and enterprise networks for security.

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6. Modem

Purpose

A modem connects a network to an Internet Service Provider (ISP).

How it Works

It converts digital data into signals suitable for transmission over communication lines.

Real-World Usage

Used in homes and businesses to provide internet access.

Part B: IP Address Classification

IP Address	Public / Private	Explanation
192.168.1.10	Private	Falls within the private range 192.168.0.0 – 192.168.255.255
10.0.0.5	Private	Falls within the private range 10.0.0.0 – 10.255.255.255
172.16.5.20	Private	Falls within the private range 172.16.0.0 – 172.31.255.255
8.8.8.8	Public	Google's public DNS server
1.1.1.1	Public	Cloudflare's public DNS server
192.168.100.1	Private	Falls within the private range 192.168.0.0 – 192.168.255.255

Part C: Understanding Your Network

Network Information

- IPv4 Address: 172.21.11.65
- Default Gateway: 172.21.11.1
- DNS Server: 172.21.3.100

Which IP range does your device belong to?

My device belongs to the 172.16.0.0 – 172.31.255.255 private IP address range.

Is it Public or Private?

It is a Private IP address because it falls within the reserved private IP range and is used within a local network.

What role does your router play in your network?

The router acts as the gateway between my local network and the internet. It forwards data packets between devices on the network and external servers.

What would happen if the DNS server stopped working?

Without DNS, domain names such as google.com could not be translated into IP addresses. Users would have to access websites using numerical IP addresses instead of domain names.

Part D: Network Communication Flow

Network Diagram

Your Device (172.21.11.65)
↓
Router / Gateway (172.21.11.1)
↓
DNS Server (172.21.3.100)
↓
Google Server (142.251.220.46)
↓
Response Back to Device

Step 1: User Request

The user enters www.google.com into a web browser.

Step 2: Router Forwarding

The router receives the request and forwards it to the configured DNS server.

Step 3: DNS Resolution

The DNS server translates the domain name google.com into an IP address.

Step 4: Communication with Google

The device sends a request to Google's server using the resolved IP address.

Step 5: Response

Google processes the request and sends the webpage data back to the user's device through the network.

Part E: Practical Command Exercise

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C:\Users\KJC Staff>nslookup google.com
Server:  UnKnown
Address:  172.21.3.100

Non-authoritative answer:
Name:    google.com
Addresses:  2404:6800:4009:808::200e
          142.251.220.46
```

```
C:\Users\KJC Staff>ping google.com

Pinging google.com [142.251.42.238] with 32 bytes of data:
Reply from 142.251.42.238: bytes=32 time=13ms TTL=118
Reply from 142.251.42.238: bytes=32 time=14ms TTL=118
Reply from 142.251.42.238: bytes=32 time=13ms TTL=118
Reply from 142.251.42.238: bytes=32 time=14ms TTL=118

Ping statistics for 142.251.42.238:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 13ms, Maximum = 14ms, Average = 13ms
```

Command Results

What IP address did DNS return for Google?

DNS returned:

IPv4 Address: 142.251.220.46

IPv6 Address: 2404:6800:4009:808::200e

Was the ping successful?

Yes. The ping was successful.

Packets Sent: 4

Packets Received: 4

Packets Lost: 0

Packet Loss: 0%

How many hops were shown?

The traceroute showed 8 hops before reaching Google's server.

Why is DNS important before communication begins?

DNS is important because computers communicate using IP addresses, not domain names. DNS translates user-friendly names such as google.com into IP addresses so devices can locate and communicate with the correct server.

Conclusion

This task helped me understand network devices, IP addressing, DNS resolution, and the flow of network communication. I also learned how to use networking commands such as ipconfig, nslookup, ping, and tracert to analyze and troubleshoot network connectivity.